

NLP Limitations with LLM Qualitative Coding

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This dataset is associated with an EMNLP 2026 Findings paper, titled “What Limits Us? Analyzing Self-Reported Limitations in NLP Research”. The dataset contains qualitative codes assigned by a large language model to the Limitations sections of 16,047 papers from the ACL Anthology published between 2020 and 2025. Every code assignment is traced to the sentences that triggered it, with a character span into the Limitations section and a short justification written by the model. It is organised paper-first: the top level is the paper, and each record carries the codes assigned to it along with the evidence supporting each one. All labels are machine-generated; the dataset carries no human annotation and is intended for large-scale analysis of how limitation statements are reported, not as a reference standard. A companion dataset of 150 papers coded by four human annotators is released separately.

Data Card

DATASET TEAM(S)	DATASET CONTACT	DATASET AUTHORS															
- Technology, AI, Society, and Culture (TASC) Team, Google - Data MIND Lab, Department of Computer Engineering, Faculty of Engineering, Chulalongkorn University	- Tawan Thaeprasit: tawanth.official@gmail.com - Peeranuth Kehasukcharoen: 6030416021@alumni.chula.ac.th	- Tawan Thaeprasit, Chulalongkorn University in Thailand - Peeranuth Kehasukcharoen, Chulalongkorn University in Thailand - Ding Wang, Google - Remi Denton, Google - Peerapon Vateekul, Chulalongkorn University in Thailand - Piyawat Lertvittayakumjorn, Google															
PRIMARY DATA MODALITY	DATASET SNAPSHOT	DESCRIPTION OF CONTENT															
Image Data Text Data Tabular Data Audio Data Video Data Time Series Graph Data Geospatial Data Multimodal Others Unknown	Size of dataset 121.3 MB (JSON, primary) / 107.4 MB (CSV) Number of Instances 16,047 papers across four files Codes in the codebook 50 Code assignments 84,930 Evidence spans 131,166 Average codes per paper 5.29 Average spans per paper 8.17 Algorithmic Labels 84,930 Human Labels 0 An assignment is one code applied to one paper. A paper carries between 1 and 19 codes, and between 1 and 62 evidence spans.	Released as four files, each in JSON (primary) and CSV, coded against the codebook version current when that year-group was processed. Codebook growth is additive: no code was removed once introduced, though three were renamed for clarity between the first two versions. File <table><tr><th>Years</th><th>Papers</th><th>Codebook version (size)</th></tr><tr><td>papers_2020_2022</td><td>2020–2022</td><td>1,595 codebook_initial (25)</td></tr><tr><td>papers_2023</td><td>2023</td><td>4,024 codebook_after_2022 (47)</td></tr><tr><td>papers_2024</td><td>2024</td><td>4,159 codebook_after_2023 (47)</td></tr><tr><td>papers_2025</td><td>2025</td><td>6,269 codebook_after_2024 (48)</td></tr></table>	Years	Papers	Codebook version (size)	papers_2020_2022	2020–2022	1,595 codebook_initial (25)	papers_2023	2023	4,024 codebook_after_2022 (47)	papers_2024	2024	4,159 codebook_after_2023 (47)	papers_2025	2025	6,269 codebook_after_2024 (48)
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DATASET SUBJECT	EXAMPLE: DATA POINT	DATA FIELDS															
Sensitive Data about people Non-Sensitive Data about people Data about natural phenomena Data about places and objects Synthetically generated data Data about systems or products Others: text published in academic proceedings Unknown	paper_id 2020.acl-main.189 title Active Imitation Learning with Noisy Guidance existing_code ['Methodological Constraints', 'Non-Limitation: Anticipated Impact', 'Non-Limitation: Method Details', 'Non-Limitation: Method Strength', 'Non-Limitation: Strong Reported Performance'] new_code [] existing_code_details[0].code_name 'Methodological Constraints' .evidence[0].segmented_id '2020.acl-main.189_s05' .evidence[0].segmented 1. In some settings, learning a difference classifier may be as hard or harder than learning the structured predictor; for instance if the task is binary sequence labeling (e.g., word segmentation), minimizing its usefulness.' .evidence[0].highlight_start/end 746 / 949	paper_id (str) ACL Anthology identifier, unique per record title (str) paper title as published abstract (str) paper abstract as published; empty for 4 of 16,047 records limitation (str) verbatim Limitations section segmented_text (str → dict[str, str]) JSON map of sentence_id to sentence, decoded with json.loads existing_code (list[str]) codes matched to the codebook version this file was coded against new_code (list[str]) codes created where no codebook entry fitted existing_code_details (list[dict]) one entry per code in existing_code: {code_name (str), evidence (list[dict])} new_code_details (list[dict]) same structure, for new_code evidence[0].segmented_id (str) key into segmented_text identifying which sentence this span falls within															

	.evidence[0].highlight_text 'learning a difference classifier may be as hard or harder than learning the structured predictor; for instance if the task is binary sequence labeling ...' .evidence[0].justification 'The authors identify a structural vulnerability in their own methodology: the internal component they rely on ... can become too difficult to learn in certain settings, undermining the method's utility.' new_code_details []	evidence[].segmented (str) the text of that sentence, copied from segmented_text[segmented_id] evidence.highlight_start, highlight_end (int) offsets into the full limitation string, not into any single sentence evidence.highlight_text (str) the highlighted span evidence.justification (str) mean 115 characters, min 12, max 419 Note: evidence entries carry no sentence_id. Linking a span back to a segmented_text entry requires matching its offset range against the sentence boundaries, since the two are not directly cross-referenced in this release.

DATASET PURPOSE(S)	KEY DOMAINS OR APPLICATION(S)
Monitoring Research Production Others	Domains Natural language processing, qualitative coding methodology, meta-research on scholarly writing. Problem Space What NLP researchers report as the limitations of their own work, at a scale no manual coding effort could reach, and how that reporting shifts across venues and years.

DATASET USAGE	INTENDED AND/OR SUITABLE USE CASE(S)
Safe for production use Safe for research use Conditional use Only approved use Others	– Large-scale analysis of which limitation categories are reported, and how that distribution moves across venues and years. – Studying rhetorical strategy through the Non-Limitation codes, which mark content that sits in a Limitations section without being a limitation. – Tracing a claim back to the sentence that produced it, using the evidence spans and justifications. Unsuitable as ground truth. These are model judgements, not human ones. Where a reference standard is needed, use the companion human-coded dataset. Also unsuitable for year-on-year comparison without weighting, given the coverage imbalance described below.

SAFETY OF USE WITH OTHER DATA	ACCEPTABLE TRANSFORMATIONS
Safe to use with other data Conditionally safe to use with other data Should not be used with other data Unknown	Joining with other datasets Subsampling and splitting Filtering Joining input sources Cleaning missing values Anomaly detection Grouping and summarizing Scaling and reducing Statistical transformations Redaction or Anonymization Join on paper_id, which resolves against ACL Anthology metadata. Pivoting the file from code-first to paper-first is the usual first step, since most analyses are per paper.

VERSION STATUS	DATASET VERSION	MAINTENANCE PLAN
Regularly Updated Actively Maintained Limited Maintenance Deprecated	Last Updated Jun 2026 Release Date Aug 2026 The codebook itself went through several versions, revised annually with humans in the loop as new material was coded. All versions are released, and the 50-code codebook documented here is the final one.	The dataset is a fixed output of one coding run. No expansion is planned, but errors will be corrected if they surface.

ACCESS POLICY	RETENTION POLICY	WIPEOUT POLICY
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Public. Released under CC BY 4.0 in the same GitHub repository as the codebook versions and the companion human-coded dataset. Repository https://github.com/Sundione/nlp-self-reported-limitations CC BY 4.0 follows ACL Anthology practice for material published from 2016 onward. Attribution to the original paper authors is required for the quoted sentences and is carried by paper_id.	No retention limit. The dataset is intended to remain available as a fixed record of the coding run, and contains no personal data.	Not applicable. No deletion schedule applies.
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DATA COLLECTION METHODS	DATA SOURCES	DATA COLLECTION
API Artificially Generated Crowdsourced – Paid Crowdsourced – Volunteer Vendor Collection Efforts Scraped or Crawled Survey, forms or polls Taken from other existing datasets Unknown	Source ACL Anthology, https://aclanthology.org/ Coverage 2020 to 2025, six venue tracks Source papers were retrieved and extracted from ACL Anthology; the codes themselves were generated by a language model, which is why both collection methods are selected.	Timeline Dec 2025 – Aug 2026 Data Modality Text Data Annotations Machine-generated by a large language model Date of coding run Apr 2026 – June 2026

INCLUSION CRITERIA	EXCLUSION CRITERIA	DATA PROCESSING
– Paper is indexed in the ACL Anthology. – Published between 2020 and 2025. – Contains an identifiable Limitations section whose text could be extracted. – Published in an ACL or EMNLP main or Findings track.	– Papers with no Limitations section, which cannot be coded.	Limitations sections were extracted from the source PDFs with the Python library docling. The extracted text was segmented into semantic units by Gemini 2.5 Flash, prompted to preserve meaning rather than to split on orthographic sentence boundaries. The output was parsed into the sentence keys used here and was not manually corrected. Each Limitations section was then coded against the codebook by a language model, which applied existing codes where they fit and, where none did, proposed new codes inductively..

SENSITIVE DATA	FIELDS WITH SENSITIVE DATA	SECURITY AND PRIVACY HANDLING
User Content User Metadata Identifiable Data S/PII Business Data Employee Data Health Data Children's Data None	No field stores personal data directly. Indirect paper_id resolves to a public paper with named authors, and the quoted sentences are that paper's own words.	The source text is already public, so no de-identification was applied and none is required.

SENSITIVE HUMAN ATTRIBUTES	SOURCE(S) OF HUMAN ATTRIBUTES	RATIONALE FOR COLLECTING HUMAN ATTRIBUTES
Race Gender Ethnicity Socio-economic status Geography Language Sexual Orientation Religion Age Culture Disability Experience or Seniority None collected	Not applicable. No human attribute was collected, labelled or derived.	Not applicable. The unit of analysis is a written statement, not a person.

TRANSFORMATIONS APPLIED	LIBRARIES AND METHODS USED
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Anomaly Detection Cleaning Mismatched Values Cleaning Missing Values Converting Data Types Data Aggregation Dimensionality Reduction Joining Input Sources Redaction or Anonymization	PDF extraction docling Segmentation Gemini 2.5 Flash, semantic units. Coding model Gemini 3.1 Pro Decoding temperature = 0 Segmentation and coding are the only transformations. No value in the released file was cleaned, imputed or type-converted after the coding run.
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SAMPLING METHOD(S)	SAMPLING CHARACTERISTIC(S)	SAMPLING CRITERIA
Cluster Sampling Haphazard Sampling Multi-stage Sampling Random Sampling Retrospective Sampling Stratified Sampling Systematic Sampling Unknown Unsampled	Population ACL Anthology, 2020 to 2025 Sample Size 16,047 papers 2020 (74) — acl-main 19, emnlp-main 37, findings-emnlp 18 2021 (84) — emnlp-main 41, acl-long 24, findings-acl 8, findings-emnlp 9, acl-short 2 2022 (1,437) — emnlp-main 817, findings-emnlp 528, acl-long 59, findings-acl 26, acl-short 7 2023 (4,024) — findings-emnlp 1,044, emnlp-main 1,042, acl-long 897, findings-acl 885, acl-short 156 2024 (4,159) — emnlp-main 1,266, findings-acl 967, findings-emnlp 997, acl-long 853, acl-short 76 2025 (6,269) — emnlp-main 1,807, acl-long 1,591, findings-emnlp 1,397, findings-acl 1,378, acl-short 96	N/A

ANNOTATION WORKFORCE TYPE	ANNOTATION CHARACTERISTICS	ANNOTATION DESCRIPTION
Machine-generated Annotations Human Annotations – Expert Human Annotations – Non-expert Human Annotations – Employees Human Annotations – Contractors Human Annotations – Crowdsourcing Human Annotations – Outsourced Unlabeled	Code assignments 84,930 Evidence spans 131,166 Codes in the codebook 50 Limitation codes 40 Non-Limitation codes 10 Codes per paper mean 5.29, range 1 to 19 Spans per paper mean 8.17, range 1 to 62 Justification length mean 115 characters, none empty	Each assignment ties a code to a paper and carries the evidence that produced it: the triggering sentence, a character span into the Limitations section, and a justification written by the model. The codebook was revised annually as coding progressed, and every version is released alongside the data. The 50-code version documented here is the final one: 40 limitation codes and 10 Non-Limitation codes marking content that appears in a Limitations section without being a limitation. Code frequencies are steeply skewed. Non-Limitation: Future Work appears in 12,041 papers and Scope Limitation in 10,091, while Sparse Data Sensitivity appears in 3 and Dataset Task Mismatch in 5, both surfaced as new_code before being formalised into codebook_after_2025.

ANNOTATOR BREAKDOWN	ANNOTATOR DESCRIPTION
Annotator type Language model Model Gemini 3.1 Pro Human annotators 0 Assignments per paper mean 5.29 Working language English	All coding was performed by a language model against the codebook. No human assigned, reviewed or adjudicated the labels in this dataset. The codebook the model worked from was developed by the research team. Its first version was built from a pilot in which an LLM inductively coded 50 Limitations sections from ACL 2024 extracted with BAGELS (Al-Azher et al., 2025); those codes were manually aligned with the taxonomy of limitations in AI research of Xu et al. (2025) and extended to cover topics the pilot surfaced. It was revised iteratively thereafter, reaching the 50 codes documented here.

VALIDATION METHOD(S)	VALIDATION BREAKDOWN	DESCRIPTION OF VALIDATION
Data Type Validation Range and Constraint Validation Code/cross-reference Validation Structured Validation	paper_id 16,047 unique, 0 duplicates Code/codebook resolution every code in every file resolves to that file's applicable codebook version	Structural checks were run programmatically over the released file: every code carries a unique identifier and name, identifiers run consecutively without gaps, every sentence key

Consistency Validation Not Validated Others: inter-annotator agreement with the model treated as a coder	Alpha, human annotators 0.644 Alpha, human annotators plus the model 0.647	matches the paper it belongs to, and no justification is empty. Coding quality was assessed by treating the model as an additional coder against the companion human-annotated dataset, a separate release of 150 papers double-coded by four human annotators, and recomputing Krippendorff's alpha, binary multi-label at paper level, over the papers that both common to both datasets. Alpha among the human annotators alone is 0.644. Adding the model as a coder gives 0.647. The model therefore does not behave as an outlier: introducing it does not degrade the reliability of the coder pool.
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VALIDATORS CHARACTERISTIC(S)	VALIDATORS DESCRIPTION(S)

ML APPLICATION(S)
A large language model produced every label in this dataset. It was not trained or fine-tuned for the task; it was prompted to apply an existing codebook and to return, for each assignment, the sentences that triggered it and a justification. No model was trained on this dataset.

Reflections on Data

These labels are model output, not ground truth	Every code in this dataset was assigned by a language model with no human verification. The justifications make each assignment inspectable, which is useful, but a fluent justification is not evidence that the assignment is correct. Where a claim needs a reference standard, the companion human-coded dataset of 150 papers exists for that purpose.
Temporal coverage	The dataset holds 74 papers from 2020 and 84 from 2021, against 6,269 from 2025. Under 1% of the data comes from the first two years. This partly reflects reality, ACL Rolling Review, the shared reviewing pipeline both ACL and EMNLP draw submissions from, began asking authors to discuss limitations when it launched in 2022, and made a dedicated Limitations section mandatory, with desk rejection for its absence, only from the December 2023 cycle onward, but it means any trend line drawn across the full range is dominated by the recent years.
Evidence spans do not always resolve against the source text	highlight_start , highlight_end and highlight_text are derived by parsing the model's own output, and the model occasionally returns a highlighted text that does not match the actual content at that position in the Limitations section, so the span cannot be resolved correctly. The defective-span rate is 1.47% overall (1,933 of 131,166). This does not affect the code assigned or its justification, only the character span. However, each evidence entry also carries segmented_id and segmented , which identify the sentence-level unit the span was drawn from independently of the character offsets; where a span fails to resolve, the segment it points to still can, and remains usable for checking the evidence behind a code.